

6a	Photon energy = $6.63 \times 10^{-34} \times 3.0 \times 10^8 / 430 \times 10^{-9}$ $= 4.63 \times 10^{-19} \text{ J}$ Number = $3.2 \times 10^{-3} / 4.63 \times 10^{-19}$ $= 6.92 \times 10^{15}$	C1 C1 A1
bi	$p = 6.63 \times 10^{-34} / 430 \times 10^{-9}$ $= 1.54 \times 10^{-27} \text{ kg m s}^{-1}$ Total momentum = $6.92 \times 10^{15} \times 1.54 \times 10^{-27}$ $= 1.07 \times 10^{-11} \text{ kg m s}^{-1}$	C1 A1
bii	Force is rate of change of momentum $F = 1.07 \times 10^{-11} \text{ kg m s}^{-2}$	B1 A1

c	Not all photons are absorbed by the surface.	B1
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